Module 2: Data Types and Toolbox — Practice

Lesson 1 | Classify, choose, summarize, interpret

Table of contents

Before you begin 1
1. Classify the variables 1
2. Explain the mistakes 1
3. Choose the toolbox 2
4. Summarize restaurant quality 2
5. Summarize audit durations 2
What to submit 2

Before you begin

Read Lesson 1. Work individually on Questions 1–3, then use R for Questions 4–5. Give reasons, units, and interpretations alongside your answers.

1. Classify the variables

For each variable, identify **categorical or numerical**, its **measurement level**, and, if numerical, **discrete or continuous**. Explain your decision. Classify the underlying quantity; mention separately if recording precision changes what is stored.

Variable	Example or definition
Restaurant identifier	101, 102, 103: labels assigned to restaurants
Payment method	Cash, card, e-wallet
Quality rating	Good, Very Good, Excellent
Number of orders	Completed orders during one hour
Waiting duration	Time from joining a queue to receiving service, recorded to the nearest minute
Temperature	Air temperature in degrees Celsius
Calendar year	Year a restaurant opened, recorded as an integer
Distance traveled	Delivery distance in kilometers

2. Explain the mistakes

1. An analyst codes quality as Good = 1, Very Good = 2, Excellent = 3 and says: “Excellent is three times as good as Good.” Explain the problem.
2. An analyst says: “The audit times have no decimal places, so elapsed audit duration must be discrete.” Explain the distinction between the quantity and its recorded values.
3. A student says: “20°C is twice as hot as 10°C.” Why does the Celsius scale not support that conclusion?

3. Choose the toolbox

Choose a suitable **table**, **chart**, and, where appropriate, **numerical measure** for each question. Explain why each choice fits the variables.

1. Which quality category occurs most often among the 300 restaurants?
2. What is a typical audit duration, and how much do durations vary?
3. How do meal-price distributions differ across quality categories?
4. In the electronics data, how are weekly advertising counts associated with sales?

Do not infer causation from an observed association.

4. Summarize restaurant quality

Download the restaurant data, or open the course R project and use the path below.

```
# read.csv(): reads a comma-separated file into a data frame; its first row supplies
 ↪  column names by default.
# c(): combines values into a vector, converting them to a common type when needed.
# factor() gives categories explicit levels and an optional order.
# The level order controls grouping and display; ordered=TRUE records ordinal category
 ↪  order.
restaurants <- read.csv("data/module02/restaurant_ratings.csv")
restaurants$quality_rating <- factor(restaurants$quality_rating,
  levels = c("Good", "Very Good", "Excellent"), ordered = TRUE)
```

1. Make a frequency table with counts and percentages. Check both totals.
2. Draw an ordered bar chart with a clear title and labeled axes.
3. State the modal category and its percentage.
4. Explain why a histogram of the codes 1, 2, 3 would hide the categorical meaning.

5. Summarize audit durations

Download the audit data, or use the course project path.

```
# names(): reads or assigns names to vector elements or table columns.
# read.csv() imports a CSV into a data frame with named columns.
# The relative path starts at the course project folder; the source CSV is not modified.
audit <- read.csv("data/module02/audit_times.csv")
names(audit) # Identify the duration column before calculating summaries.
```

1. Construct a grouped frequency table using intervals **10–<15, 15–<20, 20–<25, 25–<30, 30–<35 days**. Check that every audit is counted once.
2. Draw a histogram using the same boundaries.
3. Calculate the mean, median, sample standard deviation, minimum, maximum, range, quartiles, and IQR. Use `quantile(x, c(.25, .75), type = 6)` for the book's quartile convention.
4. Explain the mean and sample standard deviation in a sentence using days. Describe the middle half using the quartiles.
5. Compare the mean and median. What feature of the observed distribution helps explain their difference? Avoid claiming a population result from these 20 observations alone.

What to submit

A short Quarto report containing your classifications, reasons, tables, charts, and interpretations. Show enough R code to reproduce your calculations. Use the shared course style: pale lavender title panels, dark text, and restrained Petra accents. Choose readable analytical colors for your charts.